```
from openpyxl import load_workbook
import matplotlib.pyplot as plt

# File path to your Excel file
file_path = '/content/Marks_secA.xlsx'

# Load the workbook and active sheet
workbook = load_workbook(filename=file_path)
sheet = workbook.active

# Extract names and marks
names = []
marks = []

for i, row in enumerate(sheet.iter_rows(min_row=2, values_only=True), start=2):
    name = row[1]  # Assuming Name is in 2nd column (B)
    mark = row[6]  # Assuming Marks Obtained is in 7th column (G)

    try:
        mark = float(mark)
        names.append(name)
        marks.append(mark)
    except (TypeError, ValueError):
        print(f"Skipping row {i} due to invalid mark: {mark}")

# Check if we have valid marks
if not marks:
    print("No valid marks found. Please check Excel column format.")
else:
    print("\nValid marks found. Proceeding with plotting...")

    # Calculate percentage (assuming marks are out of 15)
    percentages = [(m / 15) * 100 for m in marks]

    # Categorize students
    greater_75 = [(names[i], percentages[i]) for i in range(len(percentages)) if percentages[i] > 75]
    between_60_75 = [(names[i], percentages[i]) for i in range(len(percentages)) if 60 <= percentages[i] <= 75]
    less_60 = [(names[i], percentages[i]) for i in range(len(percentages)) if percentages[i] < 60]

    # Print categorized students
    print("\n>75% Students:")
    for n, p in greater_75:
        print(f"{n}: {p:.2f}%")

    print("\n60%-75% Students:")
    for n, p in between_60_75:
        print(f"{n}: {p:.2f}%")

    print("\n<60% Students:")
    for n, p in less_60:
        print(f"{n}: {p:.2f}%")

    # Plot Histogram
    plt.figure(figsize=(10, 5))
    plt.hist(percentages, bins=range(0, 101, 5), edgecolor='black', color='skyblue')
    plt.xlabel("Percentage")
    plt.ylabel("Number of Students")
    plt.title("Histogram of Student Percentages")
    plt.grid(True)
    plt.show()

    # Plot Scatter with improved x-axis formatting
    plt.figure(figsize=(14, 6))
    plt.scatter(names, percentages, color='blue', edgecolors='black')

    plt.xlabel("Students")
    plt.ylabel("Percentage")
    plt.title("Scatter Plot of Student Percentages")

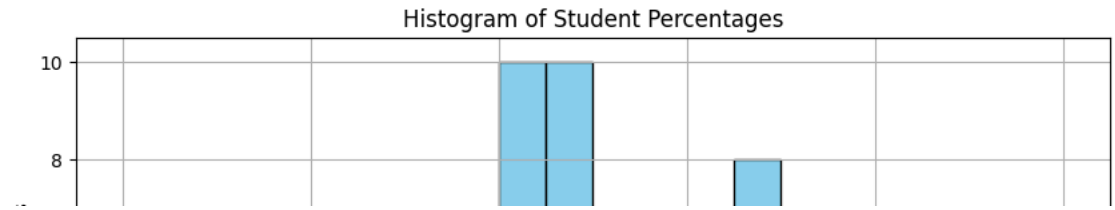
    plt.xticks(rotation=45, ha='right', fontsize=8)
    plt.tight_layout()
    plt.grid(True)
    plt.show()
```

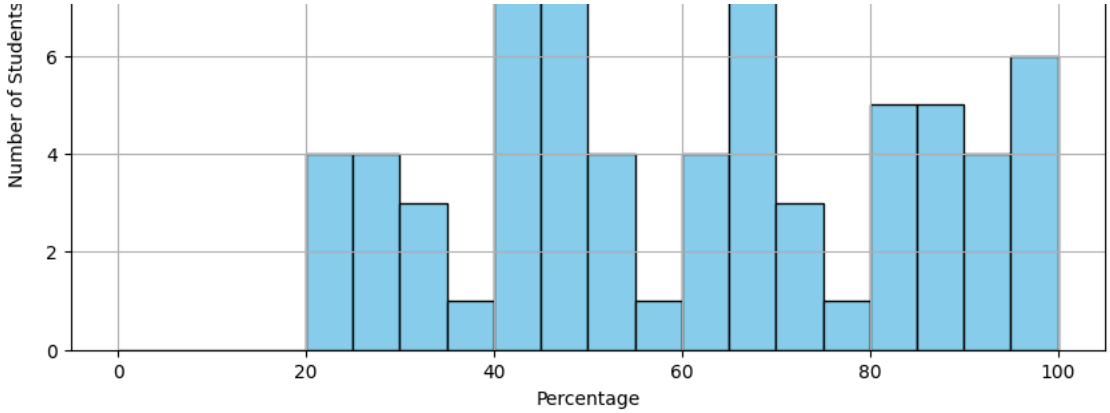
```
Valid marks found. Proceeding with plotting...

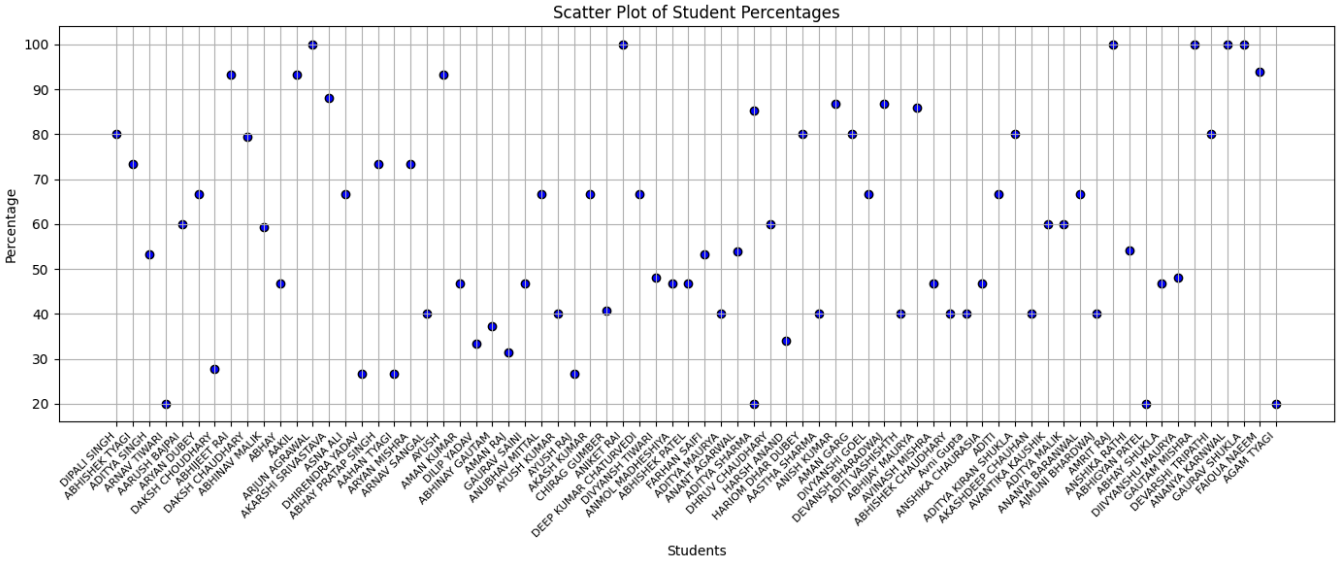
>75% Students:
DIPALI SINGH: 80.00%
ABHIJEET RAI: 93.33%
DAKSH CHAUDHARY: 79.33%
AAKIL: 93.33%
ARJUN AGRAWAL: 100.00%
AKARSHI SRIVASTAVA: 88.00%
AYUSH: 93.33%
ANIKET RAI: 100.00%
ADITYA SHARMA: 85.33%
HARIOM DHAR DUBEY: 80.00%
ANISH KUMAR: 86.67%
AMAN GARG: 80.00%
DEVANSH BHARADWAJ: 86.67%
ABHIJAY MAURYA: 86.00%
ADITYA KIRAN SHUKLA: 80.00%
AMRIT RAJ: 100.00%
GAUTAM MISHRA: 100.00%
DEVARSHI TRIPATHI: 80.00%
ANANYA KARNWAL: 100.00%
GAURAV SHUKLA: 100.00%
FAIQUA NAEEM: 94.00%

60%-75% Students:
ABHISHEK TYAGI: 73.33%
AARUSH BAJPAI: 60.00%
ARYAN DUBEY: 66.67%
ASNA ALI: 66.67%
ABHAY PRATAP SINGH: 73.33%
ARYAN MISHRA: 73.33%
ANUBHAV MITTAL: 66.67%
AKASH KUMAR: 66.67%
DEEP KUMAR CHATURVEDI: 66.67%
DHRUV CHAUDHARY: 60.00%
DIVYANSHI GOEL: 66.67%
ADITI: 66.67%
AVANTIKA KAUSHIK: 60.00%
ADITYA MALIK: 60.00%
ANANYA BARANWAL: 66.67%

<60% Students:
ADITYA SINGH: 53.33%
ARNAV TIWARI: 20.00%
DAKSH CHOUDHARY: 27.73%
ABHINAV MALIK: 59.33%
ABHAY: 46.67%
DHIRENDRA YADAV: 26.67%
AAHAN TYAGI: 26.67%
ARNAV SANGAL: 40.00%
AMAN KUMAR: 46.67%
DILIP YADAV: 33.33%
ABHINAY GAUTAM: 37.33%
AMAN RAJ: 31.33%
GAURAV SAINI: 46.67%
AYUSH KUMAR: 40.00%
AYUSH RAJ: 26.67%
CHIRAG GUMBER: 40.67%
DIVYANSH TIWARI: 48.00%
ANMOL MADHESHIYA: 46.67%
ABHISHEK PATEL: 46.67%
FARHAN SAIFI: 53.33%
ADITYA MAURYA: 40.00%
ANANT AGARWAL: 53.87%
ADITYA SHARMA: 20.00%
HARSH ANAND: 34.00%
AASTHA SHARMA: 40.00%
ADITI VASHISHTH: 40.00%
AVINASH MISHRA: 46.67%
ABHISHEK CHAUDHARY: 40.00%
Avni Gupta: 40.00%
ANSHIKA CHAURASIA: 46.67%
AKASHDEEP CHAUHAN: 40.00%
AJMUNI BHARDWAJ: 40.00%
ANSHIKA RATHI: 54.13%
ABHIGYAN PATEL: 20.00%
ABHAY SHUKLA: 46.67%
DIIVYANSHU MAURYA: 48.00%
AGAM TYAGI: 20.00%
```




Number of Students
Percentage
0
2
4
6
0
20
40
60
80
100


Scatter Plot of Student Percentages
Percentage
Students
20
30
40
50
60
70
80
90
100